

CoupletEval: A Novel Benchmark for Assessing Chinese Linguistic Proficiency in Large Language Models

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Abstract—The evaluation of linguistic capabilities in large language models (LLMs) has long lacked rigorous, unified methodologies that address both intrinsic language properties and human interpretability. To bridge this gap, we propose a novel framework that leverages Chinese couplets—a traditional literary form with strict prosodic and semantic constraints—as a challenging benchmark. Our methodology introduces both formal metrics (length match, tonal pattern opposition, part-of-speech correspondence, structural alignment, and character non-repetition) and content metrics (rhythmic harmony, semantic relevance, and imagery correspondence), enabling multidimensional assessment. A key innovation is the integration of a model self-evaluation mechanism, allowing LLMs to assess their own outputs and providing valuable insights into their metacognitive abilities. Systematic experiments with contemporary models—including Qwen3, Gemma3, DeepSeek-R1, and DeepCoder—across various parameter scales (0.6B to 7B) empirically validate the proposed approach. Results show strong concordance between model self-evaluation and conventional computational metrics, particularly in scenarios requiring deep cultural contextualization and strict formal constraints. These methodological advancements not only support computational model optimization but also promote the development of culturally informed artificial intelligence, establishing Chinese couplets as a robust benchmark for evaluating advanced language understanding and generation capabilities.

Keywords—Large Language Models, Chinese Couplets, Model Self-evaluation, AI Benchmark

I. INTRODUCTION

Couplets represent a highly regulated literary form in Chinese culture, characterized by strict requirements. Each pair of lines must not only contain the same number of characters, but also exhibit parallelism in parts of speech, structure, rhythm, content, and tonal contrast. As a unique literary art form of the Chinese nation, couplets originated in the Wei and Jin dynasties, were refined during the Tang and Song, and flourished in the Ming and Qing, maintaining a profound popular foundation and distinctive national characteristics to this day [1]. They have permeated daily life and social

activities, serving as both a scholarly pastime and a practical element in festivals, weddings, and funerals. As a carrier of the Chinese language and culture that embodies a long-standing historical tradition in form, prosody, meaning, grammar, and rhetoric, couplets entered the compulsory education stage for Chinese youth at the beginning of this century, becoming an important component of indigenous cultural ideology. Research on couplets is of significant cultural and practical value for promoting national culture and exploring the essential characteristics of the Chinese language [2], [3], [4].

With advances in computational linguistics, various natural language processing techniques have been applied to Chinese language research, including word sense disambiguation, syntactic transformation, terminology mapping, and text mining [5], [6], [7], [8], [9], [10], [11], [12]. In couplet research, methods have evolved from statistical and rule-based response models [13] to neural network-based generation [14], and further to the application in poetry composition and terminology databases [15], [16], [17]. The introduction of the attention mechanism and Transformer architectures, such as those in the GPT series, has further advanced couplet generation technology [18], [19], [20].

Meanwhile, the rapid development of large language models has expanded application scenarios, but also raised questions about their true linguistic competence. Studies have shown that LLM-generated content often diverges from human language, exhibiting issues such as hallucinations [21], [22]. To address these challenges, researchers have explored retrieval augmentation and human-in-the-loop optimization [23], [24], [25]. However, valuating LLMs’ abilities in Chinese-specific genres—especially those with deep cultural meaning and strict formal constraints—remains a significant challenge.

Although diverse evaluation frameworks have been established in areas such as standardized testing [26], [27] mathematical reasoning [28], [29] code generation [30], and model self-evaluation and reflection [31], [32], most of these works focus on overall task scores. They often overlook fine-grained linguistic abilities such as Chinese prosody, rhetoric,

and cultural imagery, and rarely apply the “self-evaluation—regeneration” paradigm to traditional Chinese literary genres.

As a key component of Chinese culture, couplets provide a unique and rigorous benchmark for assessing the capabilities of large language models in Chinese. Their creation requires not only mastery of tonal patterns and semantics, but also a deep understanding of parallelism and cultural context. Leveraging these characteristics, this study employs traditional computational linguistic methods to systematically evaluate and compare the performance of LLMs in Chinese couplet generation.

II. EVALUATION PARADIGM BASED ON CHINESE COUPLETS

A. Feasibility of Couplet-Based Evaluation

Using couplets as a tool for evaluating large language models is uniquely feasible, as couplets possess explicit and quantifiable prosodic requirements—including strict norms for character count, tonal patterns, and parallelism—which provide objective standards for assessment. Moreover, the creation of couplets demands multiple levels of linguistic competence, ranging from basic phonological and morphological processing to advanced semantic understanding and cultural interpretation, thus enabling a comprehensive evaluation of a model’s language processing abilities. The extensive historical accumulation of high-quality couplet works further offers abundant reference samples for benchmarking.

The parallel structure of couplets offers an ideal framework for testing contextual understanding and generation, supporting quantifiable, multi-dimensional assessment of both form and content. This approach allows for adjustable difficulty and directly measures cultural comprehension. As a result, couplet-based evaluation not only tests technical mastery of language rules but also deeper semantic and cultural understanding.

Given the complexity of such tasks, a single metric cannot capture all aspects of model performance. Even advanced models show significant variation when handling highly structured literary forms, reflecting differences in both linguistic proficiency and cultural awareness. Therefore, this study adopts a semi-quantitative, multi-dimensional evaluation framework, treating each metric as a relative reference. By introducing probabilistic reasoning and uncertainty, this methodology moves beyond binary evaluation and provides a more scientific foundation for assessing AI in culturally specific contexts. From an interdisciplinary perspective, this approach points toward future AI evaluation methods that emphasize multidimensional, context-aware capability profiles. The parallel structure of couplets offers an ideal framework for testing contextual understanding and generation, supporting quantifiable, multi-dimensional assessment of both form and content. This approach allows for adjustable difficulty and directly measures cultural comprehension. As a result, couplet-based evaluation not only tests technical mastery of language rules but also deeper semantic and cultural understanding.

B. Evaluation Metrics System

To comprehensively evaluate the couplet generation capabilities of large language models, this study establishes a

multidimensional quantitative metrics system, which is mainly divided into two categories: formal metrics and content metrics.

1) Pattern Metrics

a) *Length Match (LM)*: Evaluates the degree to which the generated content matches the expected length, using the formula:

$$LM(g, e) = \max(0, 1 - \frac{|len(g) - len(e)|}{len(e)}) \quad (1)$$

where g denotes the generated text, e denotes the expected text, and $len()$ represents the text length.

b) *Tone Match (TM)*: Assesses the degree to which the tones of the upper and lower lines are in opposition, using the formula:

$$TM(i, g) = \frac{\sum_{j=1}^{\min(|I|, |G|)} \mathbb{I}[(T_{i,j} = \text{"平"} \wedge T_{g,j} = \text{"仄"}) \vee (T_{i,j} = \text{"仄"} \wedge T_{g,j} = \text{"平"})]}{\min(|I|, |G|)} \quad (2)$$

c) *Part-of-Speech Match (POS)*: Evaluates the degree of part-of-speech correspondence at each position between the upper and lower lines, using the formula:

$$POS(i, g) = \frac{\sum_{j=1}^{\min(|I|, |G|)} \mathbb{I}(I_j = G_j)}{\min(|I|, |G|)} \quad (3)$$

where I and G are the part-of-speech sequences of the input and generated texts, and $\mathbb{I}()$ is the indicator function.

d) *Structure Match (SM)*: Assesses the structural symmetry between the upper and lower lines, especially the correspondence of punctuation positions, using the formula:

$$SM(e, g) = \begin{cases} 1.0, & \text{if } P_e = P_g = \emptyset \\ 0.0, & \text{if } P_e = \emptyset \text{ OR } P_g = \emptyset \\ \frac{|P_e \cap P_g|}{\max(|P_e|, |P_g|)}, & \text{Other Satuation} \end{cases} \quad (4)$$

where P_e and P_g are the sets of punctuation positions in the expected and generated texts, respectively.

e) *No Character Repetition (NC)*: Evaluates the degree to which character repetition is avoided between the upper and lower lines, using the formula:

$$NC(i, g) = 1 - \frac{|\{c | c \in g \wedge c \in i\}|}{|g|} \quad (5)$$

where $|\{c | c \in g \wedge c \in i\}|$ denotes the number of characters that appear in both the generated and input texts.

2) Content Metrics

a) *Rhythmic Harmony (RM)*: Evaluates the degree of matching between the upper and lower lines in terms of word segmentation and rhythmic patterns, using the formula:

$$RM(e, g) = \frac{\sum_{j=1}^{\min(|L_e|, |L_g|)} \mathbb{I}(|W_{e,j}| = |W_{g,j}|)}{\min(|L_e|, |L_g|)} \quad (6)$$

where L_e and L_g are the word segmentation results of the expected and generated texts, respectively, and $W_{e,j}$ and $W_{g,j}$ denote the j -th word in each.

b) Content Relevance (CR): Assesses the thematic and content-related relevance between the upper and lower lines, using the composite formula:

$$CR(i, g) = 0.2 \times S_{char}(i, g) + 0.5 \times S_{word}(i, g) + 0.3 \times S_{topic}(i, g) \quad (7)$$

where S_{char} is the character-level TF-IDF similarity, S_{word} is the word-level TF-IDF similarity, and S_{topic} is the topic similarity based on keyword extraction.

c) Imagery Correspondence (IC): Evaluates the correspondence of imagery expression between the upper and lower lines, using the formula:

$$IC(i, e, g) = \frac{IC_{count}(e, g) + IC_{reflect}(i, g)}{2} \quad (8)$$

where IC_{count} denotes the degree of noun quantity matching, and $IC_{reflect}$ denotes the degree to which nouns in the input text are reflected in the generated text.

d) LLM Self-Evaluation Metric: Utilizes the model's own cultural understanding for self-assessment, using the formula:

$$Score_{LLM}(i, g, model) = \text{Normalize}(LLM_{eval}(i, g, prompt)) \quad (9)$$

where LLM_{eval} is the model's evaluation of the generated couplet based on a specific prompt, and Normalize is the function that normalizes the original score to the $[0, 1]$ interval.

3) Comprehensive Scoring:

This study does not adopt a simple weighted average approach; instead, it retains the original scores for each dimension and constructs a capability map for the models in a multidimensional space, thereby providing a more comprehensive evaluation perspective:

$$EvaluationResult = \{LM, TM, POS, SM, NC, RM, CR, IC, Score_{LLM}\} \quad (10)$$

This multidimensional presentation method can visually display the differences in various aspects of model capabilities through radar charts, thus avoiding the evaluation bias that may arise from a single metric. For scenarios requiring ranking and comparison, weight combinations can be selectively designed according to the specific application context:

$$Score_{total} = \sum_i w_i \times Metric_i \quad (11)$$

where w_i is the weight of the i -th metric, which can be adjusted according to the evaluation focus and application requirements.

C. Implementation and Strategies

1) Evaluation Dataset Construction

This research utilized "couplet-dataset," a comprehensive evaluation corpus extracted from the "Feng Chongpu_Liwei Zhai Scattered Pages" blog containing over 700,000 couplet instances spanning diverse historical periods, cultural backgrounds, and stylistic characteristics. To ensure data quality and representativeness, the dataset underwent a series of rigorous preprocessing steps: redundant symbols and abnormal characters were removed during data cleaning; structure standardization was performed to unify separation methods and storage formats; duplicate detection algorithms eliminated identical or highly similar samples; and validity verification

ensured compliance with tonal patterns and parallelism requirements. After processing, the dataset was divided into separate files for upper lines (in.txt) and lower lines (out.txt), creating a representative corpus that comprehensively reflecting models' couplet generation and understanding capabilities across varying difficulty levels.

2) Evaluation Method Design

This research utilized "couplet-dataset," a comprehensive evaluation corpus extracted from the "Feng Chongpu_Liwei Zhai Scattered Pages" blog containing over 700,000 couplet instances spa

a) Bidirectional Generation Test: Randomly deciding whether to have the model generate the upper line based on the lower line or generate the lower line based on the upper line, ensuring fair assessment of the model's bidirectional understanding and generation capabilities.

b) Multidimensional Quantitative Scoring: Precisely quantifying the generated results across the following eight dimensions shown in formulas mentioned above. (1, 2, 3, 4, 5, 6, 7, 8)

c) Innovative Model Self-Evaluation Mechanism:

Based on the model's understanding of Chinese cultural rules, a self-assessment mechanism was designed for six core dimensions: part-of-speech correspondence, structure matching, rhythmic harmony, tonal pattern opposition, content relevance, and imagery correspondence. Models not only generate couplets but also score their own outputs on a 0-1 scale, enabling analysis of metacognitive abilities and cultural understanding by comparing algorithmic and self-evaluation scores.

This methodology integrates multidimensional metric assessment with model self-evaluation, offering several advantages: it combines objective scoring with the model's cultural knowledge, reduces human bias compared to expert-based evaluation, and allows efficient large-scale comparisons. By analyzing discrepancies between algorithmic and self-evaluation scores, the approach reveals models' meta-cognitive awareness and mastery of domain-specific knowledge. Furthermore, this framework can be extended to other culture-specific tasks, providing a systematic solution for evaluating large language models in multicultural contexts.

3) Model Comparison Strategy

To ensure fairness and comprehensiveness of the evaluation, the following strategies were employed:

a) Standardized Inputs: Unifying the format and content of input prompts across all models.

Multi-round Testing: Conducting multiple generation attempts for each model on each test item, taking average performance.

b) Cross-validation: Implementing K-fold cross-validation methodology to reduce dataset bias.

c) Gradient Analysis: Analyzing each model's capability ceiling based on performance across different difficulty levels.

d) Failure Case Analysis: Detailed documentation and analysis of typical failure cases for each model.

D. Cross-Cultural Applicability of the Evaluation Framework

This research introduces a novel evaluation methodology that combines quantitative metrics from traditional NLP approaches with model self-evaluation, creating a framework that is fundamentally language-agnostic in its design. While initially applied to Chinese couplets, this combinatorial approach can be transferred to other cultural contexts by simply adapting the specific metrics to accommodate the linguistic features and conventions of the target language, without altering the underlying evaluation architecture.

The framework's flexibility allows for seamless application across diverse literary traditions with structured forms. Japanese haiku and tanka with their syllabic requirements and seasonal references, Arabic qasida and ghazal with their specific rhyme schemes, Western sonnets and villanelles with their structural patterns—all can be evaluated through appropriate adaptations of our Length Match, Rhythmic Harmony, Imagery Correspondence, and other metrics. This adaptability demonstrates how computational evaluation can bridge linguistic boundaries while preserving sensitivity to cultural nuances.

Future research could extend this methodology to even more diverse contexts, including Greek epics, Egyptian hieroglyphic texts, and other historical literary forms across cultures. Such applications would not only advance language model evaluation techniques but also contribute to digital

of performance metrics such as response time and resource utilization, which provides valuable references for model applicability in practical scenarios

B. Models Selection

To comprehensively assess performance on Chinese culture-specific tasks, we selected representative LLMs from the open-source community, considering diversity in architecture (e.g., Token Free, mixture-of-experts, reinforcement learning, lightweight Transformers), parameter scale (0.6B–7B), Chinese language proficiency, and cultural knowledge reserves. The evaluated models—Qwen3, Gemma3, DeepSeek-R1, and DeepCoder—cover a range of technical backgrounds and training strategies (see Table 1).

The selected models differ in parameter scale, Chinese language proficiency, cultural knowledge, and architectural design, which directly impact their performance in couplet generation. For example, larger models such as Qwen3-4B and DeepSeek-R1-7B demonstrate superior format control and content richness, especially in handling complex structures and maintaining prosodic rules, while smaller models like Qwen3-0.6B and DeepCoder-1.5B are more efficient but often struggle with cultural nuance and structural complexity.

Models primarily trained on Chinese data (Qwen3, DeepSeek-R1) consistently outperform multilingual models (Gemma3) in tasks requiring deep understanding of Chinese prosody and cultural references. Qwen3 and DeepSeek-R1, with extensive Chinese corpora and cultural knowledge, generate couplets with richer imagery and more accurate parallelism. In contrast, DeepCoder, which relies on general and code-mixed corpora, tends to produce outputs lacking in cultural depth and stylistic conformity.

From a technical perspective, Qwen3's TokenFree architecture and improved RoPE enhance its ability to model long-range dependencies and tonal patterns. DeepSeek-R1's mixture-of-experts and reinforcement learning optimization contribute to its robustness in generating structurally complex and semantically coherent couplets. Gemma3's multilingual optimization, while improving generalization, sometimes reduces sensitivity to Chinese-specific linguistic features. DeepCoder's lightweight Transformer design favors inference efficiency but limits its expressive power in culture-specific tasks.

For instance, in the Length Match and No Character Repetition metrics, Qwen3-4B and DeepSeek-R1-7B approach human-level performance, while DeepCoder-1.5B frequently exhibits character repetition and format errors. In content relevance and imagery correspondence, models with richer cultural knowledge (Qwen3, DeepSeek-R1) produce more contextually appropriate and vivid couplets, (see Section X for detailed results and examples).

Table 1 Selected Models and Characteristics

Model Series	Parameter Scale	Chinese Capability	Cultural Knowledge
Qwen3	0.6B/1.7B/4B	Excellent	Rich
Gemma3	1B/4B	Good	Average
DeepSeek-R1	1.5B/7B	Outstanding	Rich
DeepCoder	1.5B	Average	Average

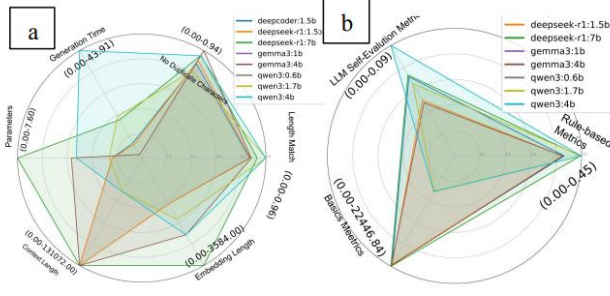
humanities by creating tools for analyzing and preserving cultural heritage. The framework thus offers both practical utility for assessing large language models and broader implications for understanding how computational systems can engage with the rich diversity of human linguistic and cultural expression.

III. EXPERRIMENT

A. Offline Execution

This study employed offline execution of all models to ensure precise regulation of computational resource allocation and parameter constraints. Offline deployment offers multiple advantages over online API calls for evaluating couplet generation capabilities:

- Keep parameters stable to avoid uncertainty from version changes.
- Control inference settings for consistent and repeatable results.
- Avoid API limits and costs, helpful for large-scale testing.
- Directly track response time and resources for practical reference. Facilitating direct observation and recording



a: comparative evaluation of model performance
b: comprehensive capabilities in couplet generation
Fig. 1. Result for Couplet Generation

These findings highlight that both model architecture and training data composition play critical roles in determining LLMs' effectiveness in Chinese couplet generation, especially for tasks requiring deep cultural and linguistic understanding.

IV. RESULTS AND ANALYSIS

The experimental results (Figure 1) reveal clear performance differences among LLMs in couplet generation. Qwen3-4B and DeepSeek-R1-7B achieve consistently strong results across all metrics, especially in format control and content richness, while Gemma3-4B and DeepCoder-1.5B lag in culture-specific tasks and structural complexity. Larger models demonstrate better adherence to prosodic and structural constraints, whereas smaller models, though more efficient, often fail to capture cultural nuance and exhibit more format errors. This highlights the persistent trade-off between model scale, efficiency, and cultural competence.

To further elucidate the underlying causes of these differences, we analyze several key factors:

- Advanced architectures and larger parameter count (e.g., Token Free, mixture-of-experts) enhance the ability to model long-range dependencies and complex linguistic patterns, directly improving performance on tasks with strict formal requirements.
- Models extensively trained on high-quality Chinese corpora (Qwen3, DeepSeek-R1) better capture the linguistic and cultural subtleties of couplets, resulting in richer imagery and more accurate parallelism. In contrast, models with general or code-mixed data (DeepCoder) lack such depth.
- Fine-tuning on culture-specific data enables models to generate contextually appropriate and stylistically conformant outputs, which is less evident in multilingual models like Gemma3.
- While multilingual models offer broader applicability, they may underperform in tasks requiring deep specialization, as seen in Gemma3's handling of Chinese couplets.

The consistency between algorithmic and self-evaluation scores reflects the models' metacognitive awareness. Larger models show higher alignment, likely due to richer internal representations and exposure to self-reflective data.

V. CONCLUSION

This study systematically compared the performance of mainstream large language models on Chinese couplet generation, leveraging traditional computational linguistic methods, a multidimensional quantitative metrics system, and an innovative model self-evaluation mechanism. Our comprehensive assessment covered prosodic features, rule adherence, content relevance, and metacognitive ability. Experimental results reveal that larger models with extensive Chinese pre-training data consistently outperform smaller models on both basic and rule-based metrics. Localized Chinese models, such as Qwen3 and DeepSeek-R1, demonstrate clear advantages in culture-specific tasks, better capturing prosodic and cultural nuances. In contrast, smaller models, while more efficient, often struggle with complex structures and cultural expression. Notably, all evaluated models exhibit limited self-evaluation (metacognitive) capability, highlighting the need for further improvement in this area.

The proposed framework provides a systematic and reproducible approach for evaluating and selecting large language models in Chinese cultural contexts. By integrating objective metrics with rigorous experimental controls, we ensure reliable and comprehensive assessment. Looking ahead, combining human evaluation with automated metrics will enable deeper exploration of models' performance in advanced areas such as artistic conception, creativity, and cultural nuance.

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